

Risk Factors for Radiation Maculopathy after Ophthalmic Plaque Radiation for Choroidal Melanoma

PAUL T. FINGER, KIMBERLY J. CHIN, AND GUO-PEI YU, ON BEHALF OF THE PALLADIUM-103 FOR CHOROIDAL MELANOMA STUDY GROUP

- **PURPOSE:** To determine how tumor characteristics and radiation dose affect the incidence of radiation maculopathy (RM).
- **DESIGN:** Retrospective, consecutive case series.
- **METHODS:** A consecutive case series of 384 uveal melanomas irradiated (mean apical dose, 71.2 Gy) were followed up for a mean 47.2 months. Tumor locations included: 122 (32%) centered anterior to the equator, 27 (7%) equatorial, and 235 (61%) posterior. Tumor sizes were American Joint Committee on Cancer class T1 (n = 180), T2 (n = 150), T3 (n = 47), and T4 (n = 7).
- **RESULTS:** RM occurred in 8 (7%) eyes with anterior uveal melanomas. In contrast, it was found in 82 (41%) eyes with posterior tumors. Multivariate analysis revealed the risk related to posterior location was greater compared with anterior location with a hazard ratio of 6.66 (95% confidence interval [CI], 4.94 to 22.50; $P = .0001$). Tumor height (> 6.0 mm) also demonstrated a high risk for RM (hazard ratio, 4.5; 95% CI, 2.68 to 10.17; $P = .0001$). A significant dose-response relationship was found between dose to fovea and RM ($P = .0005$, for trend). As compared with a dose of < 35 Gy, the risk of RM was 1.74 (95% CI, 0.98 to 3.1) for doses from 35 to 70 Gy, and the risk of RM was 2.43 (95% CI, 1.48 to 4.0) for doses of 70 Gy or more. Of interest, those anterior melanomas with RM had a mean apical height of 9.4 mm, as compared with a mean height of 3.3 mm for anterior tumors not associated with RM. Visual acuity was preserved if the fovea dose was less than 35 Gy.
- **CONCLUSIONS:** This study suggests that tumor location, tumor thickness, and radiation dose to the fovea are risk factors for the development of RM. (Am J Ophthalmol 2010;149:608–615. © 2010 by Elsevier Inc. All rights reserved.)

RADIATION ONCOLOGISTS STRIVE TO FOCUS DOSE within the targeted zone with relative sparing of normal tissues.^{1–8} However, all forms of ophthalmic irradiation risk inducing cataract, radiation retinopathy, and optic neuropathy.^{7,9–11}

Accepted for publication November 1, 2009.

From The New York Eye Cancer Center, New York, New York (P.T.F., K.J.C.), and The New York Eye and Ear Infirmary, New York, New York (G.-P.Y., P.T.F.).

Inquiries to Paul T. Finger, MD, The New York Eye Cancer Center, 115 East 61 Street, Suite 5B, New York, NY 10065; pfinger@eyecancer.com

Ophthalmic plaque radiation therapy involves either affixing radiation seeds (radionuclides) within a gold seed carrier or purchasing the seeds as a solid metal plaque encasing the source.¹² These plaques are sewn onto the sclera as to cover the tumors base (plus a 2- to 3-mm margin of normal-appearing tissue).¹³ Although the gold will block more than 99.75% of ¹²⁵I or ¹⁰³Pd irradiation, the episcleral aperture allows radiation to enter the eye.⁸

Once inside the eye, radiation dose deposition is inversely related to the square of the distance from the source, photon energies, and absorption characteristics of exposed tissues. For example, although both are equally affected by the inverse square law, the lower energy photons emitted from a ¹⁰³Pd seed (versus ¹²⁵I) are absorbed more rapidly in tissue and are less likely to reach most normal ocular structures.¹⁴ In so far as preoperative comparative dosimetry (¹²⁵I, ¹⁰³Pd, and ¹⁰⁶Ru) can be performed to examine the dose to the tumor and normal ocular structures (fovea, optic disc, lens, subjacent sclera, and opposite eye wall), comparative dosimetry can be used to choose and thus reduce dose to selected normal ocular structures.

We examined the basic premise that decreased radiation to the fovea will diminish the incidence of radiation maculopathy (RM). Further, we evaluated tumor size, location, and radiation dose to the fovea as biomarkers (risk factors) for RM. This is important because RM is the most common irreversible cause of radiation-related loss of vision.

METHODS

WE REPORT ON 384 CONSECUTIVE PATIENTS WITH UVEAL melanomas treated with ¹⁰³Pd. Our methods of diagnosis, informed consent, treatment, and follow-up have been described.¹⁴ Aspects of particular importance to this study include that tumor basal dimensions were determined by ophthalmoscopy, transillumination, fluorescein angiography, and ultrasonography.

- **TUMOR CHARACTERISTICS:** The categories of tumors were grouped according to the location of the tumor apex. Tumors grouped as posterior were centered: posterior, posterior equator, and equator posterior.¹⁵ The categories of anterior intraocular tumors included: equator anterior, ciliary body, iridociliary, and iris.¹⁵ One hundred twenty-